RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 8A: Graphing Techniques

1 PJC Prelim 9758/2018/01/Q1

It is given that $\mathrm{f}(x) = 5x^3 + 7x^2 - kx + 2$, where k is a constant.

(i) If the graph of $y = \mathrm{f}(x)$ is strictly increasing, find the range of values of k. [3]

(ii) If $k = 0$, state the range of values of x where the graph of $y = \mathrm{f}(x)$ is concave downward. [1]

1(i)	$\mathrm{f}(x) = 5x^3 + 7x^2 - kx + 2$ $\mathrm{f}'(x) = 15x^2 + 14x - k$ f is strictly increasing, so we consider $\mathrm{f}'(x) > 0$ for all x $\Rightarrow 15x^2 + 14x - k > 0$ for all x $\Rightarrow$ Discriminant < 0 , $(14)^2 - 4(15)(-k) < 0$ $k < -\frac{49}{15}$ or $k < -3.27$
(ii)	$k = 0 \Rightarrow \mathrm{f}(x) = 5x^3 + 7x^2 + 2$ For graph of $y = \mathrm{f}(x)$ to be concave downward, $\mathrm{f}''(x) = 30x + 14 < 0$ $\Rightarrow x < -\frac{7}{15}$ or $x < -0.467$

2 NJC Promo 9758/2020/Q4

A curve has equation $y + \ln y = \mathrm{e}^x$.

(i) Show that $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(y^2 + y\right) = y^2\mathrm{e}^x + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2$. [2]

(ii) Explain why the curve is concave upwards for all real values of x. [2]

2 (i)	$y + \ln y = \mathrm{e}^x$ $\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\frac{\mathrm{d}y}{\mathrm{d}x}}{y} = \mathrm{e}^x$

	$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}+\dfrac{\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}y-\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2}{y^2}=\mathrm{e}^x$ $y^2\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}+\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}y-\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2=y^2\mathrm{e}^x$ $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(y^2+y\right)=y^2\mathrm{e}^x+\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2$ (shown)
(ii)	$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(y^2+y\right)=y^2\mathrm{e}^x+\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2$ Since $\ln y$ is defined, $y>0$, and $\mathrm{e}^x>0$ for all $x\in\mathbb{R}$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{y^2\mathrm{e}^x+\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2}{y^2+y}>0$ for $x\in\mathbb{R}$. The curve is concave upwards for $x\in\mathbb{R}$.

3 Show that $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(4x+\sin x-8\sin\dfrac{x}{2}\right)=2\cos^2\dfrac{x}{2}-4\cos\dfrac{x}{2}+3$. (Hint: $\cos 2\theta=2\cos^2\theta-1$)

Hence, by completing the square, show that $4x+\sin x-8\sin\dfrac{x}{2}$ increases as x increases for all real values of x. Deduce that $4x+\sin x-8\sin\dfrac{x}{2}>0$ for $x>0$. **[5]**

3	$\text{LHS}=\dfrac{\mathrm{d}}{\mathrm{d}x}\left(4x+\sin x-8\sin\dfrac{x}{2}\right)$ $=4+\cos x-8(\dfrac{1}{2})\cos\dfrac{x}{2}$ $=4+(2\cos^2\dfrac{x}{2}-1)-8(\dfrac{1}{2})\cos\dfrac{x}{2}$ $=2\cos^2\dfrac{x}{2}-4\cos\dfrac{x}{2}+3=\text{RHS}$ (Shown)
	$2\cos^2\dfrac{x}{2}-4\cos\dfrac{x}{2}+3=2(\cos^2\dfrac{x}{2}-2\cos\dfrac{x}{2})+3$ $=2(\cos^2\dfrac{x}{2}-2\cos\dfrac{x}{2}+1-1)+3$ $=2(\cos\dfrac{x}{2}-1)^2+1$ $\because 2(\cos\dfrac{x}{2}-1)^2+1\geq 1>0\quad\forall x\in\mathbb{R}$, i.e. the gradient of $4x+\sin x-8\sin\dfrac{x}{2}$ is always positive $\forall x\in\mathbb{R}$, $\therefore 4x+\sin x-8\sin\dfrac{x}{2}$ increases as x increases for all real values of x. (Shown)

When $x=0$, $4x+\sin x-8\sin\frac{x}{2}=0+0+0=0$

And $4x+\sin x-8\sin\frac{x}{2}$ increases as x increases for all real values of x

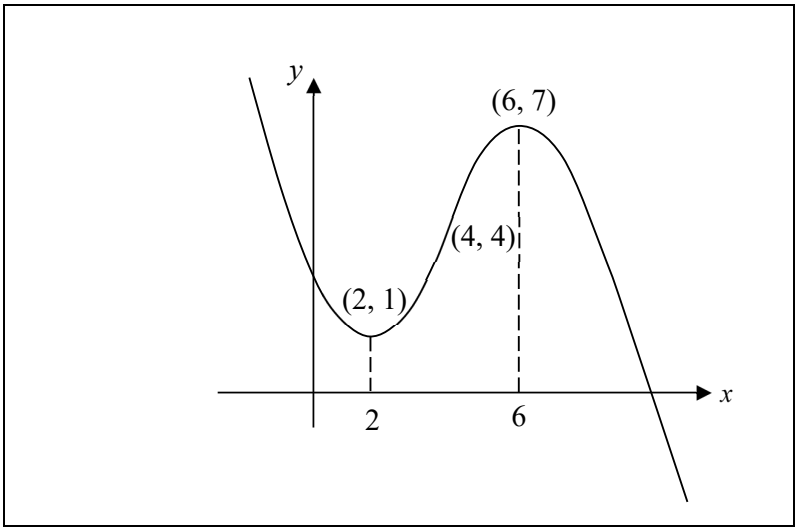
$\therefore 4x+\sin x-8\sin\frac{x}{2}>0$ for $x>0$.

4 RJC JC1 CT 9740/2006/01/Q8

Sketch a continuous curve $y=\mathrm{f}(x)$ with the following properties, labelling the coordinates where possible.

x	y	Behaviour
$x<2$		Strictly decreasing, Concave upwards
2	1	Stationary point
$2<x<4$		Strictly increasing, Concave upwards
4	4	Point of inflexion
$4<x<6$		Strictly increasing, Concave downwards
6	7	Stationary point
$x>6$		Strictly decreasing, Concave downwards

[3]



Rational Functions

5 JPJC Promo 9758/2020/Q4

The curve C has equation $y=\dfrac{2x+a}{x+b}$, for non-zero constants a and b. The graph of C passes through the point with coordinates $(0,1)$. The line $x=6$ is an asymptote to the graph of C.

(i) Determine the values of a and b. [2]

(ii) Sketch the graph of C, clearly stating the equations of the asymptotes and the coordinates of the axial intercepts. [3]

(iii) By adding a suitable graph to your sketch in **(ii)**, deduce the number of real roots for $2x^3+ax^2+x+b=0$. [3]

(i)

$y=\dfrac{2x+a}{x+b}=2+\dfrac{a-2b}{x+b}$

Since $x=6$ is an asymptote,

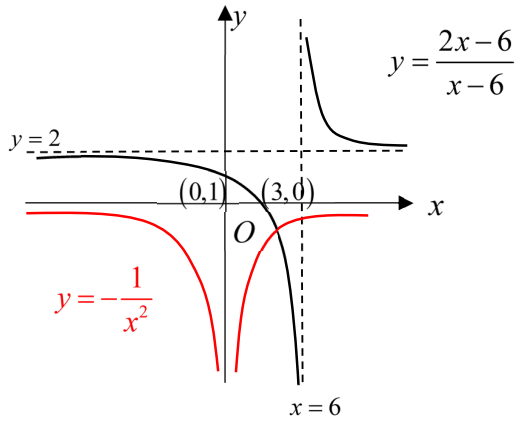
$b=-6$

Graph passes through $(0,1)$

$1=\dfrac{a}{-6}$

$a=-6$

(ii)



(iii)

$2x^3+ax^2+x+b=0$

$x^2(2x+a)=-x-b$

$\dfrac{2x+a}{x+b}=-\dfrac{1}{x^2}$

Since the graphs of $y=\dfrac{2x+a}{x+b}$ and $y=-\dfrac{1}{x^2}$ intersect only once, there is 1 real root for $2x^3+ax^2+x+b=0$.

6 RVHS JC1 Promo 9758/2019/Q8

The curve C has equation $y = \dfrac{x^2 + ax + 6}{x - 2}$, where a is a constant. It is given that the line $y = x + 5$ is an asymptote of C.

(i) Show that $a = 3$. [2]

(ii) Prove algebraically that y cannot lie between -1 and 15. [2]

(iii) Sketch the curve C, stating clearly the coordinates of any points of intersection with the axes, the coordinates of any stationary points and the equations of asymptotes. [3]

(iv) Hence, solve the inequality $\dfrac{x^2 + 3x + 6}{x - 2} \geq -x - 3$. [2]

(i)

$$y = \frac{x^2 + ax + 6}{x - 2} = x + 5 + \frac{k}{x - 2}, \text{ for some } k \in \mathbb{R}$$

$$= \frac{x^2 + 3x + (k - 10)}{x - 2}$$

$x^2 + ax + 6 = x^2 + 3x + (k - 10)$

By comparing coefficients of x, $a = 3$ (shown)

Alternatively,

$$y = \frac{x^2 + ax + 6}{x - 2}$$

$= x + (a + 2) + \dfrac{2a + 10}{x - 2}$ (performing long division)

Hence, $a + 2 = 5 \Rightarrow a = 3$ (shown)

(ii)

$$y = \frac{x^2 + 3x + 6}{x - 2}$$

$xy - 2y = x^2 + 3x + 6$

$x^2 + (3 - y)x + (2y + 6) = 0$

For this quadratic equation to have real roots, discriminant ≥ 0

i.e. $(3 - y)^2 - 4(2y + 6) \geq 0$

$\Rightarrow y^2 - 14y - 15 \geq 0$

$\Rightarrow (y + 1)(y - 15) \geq 0$

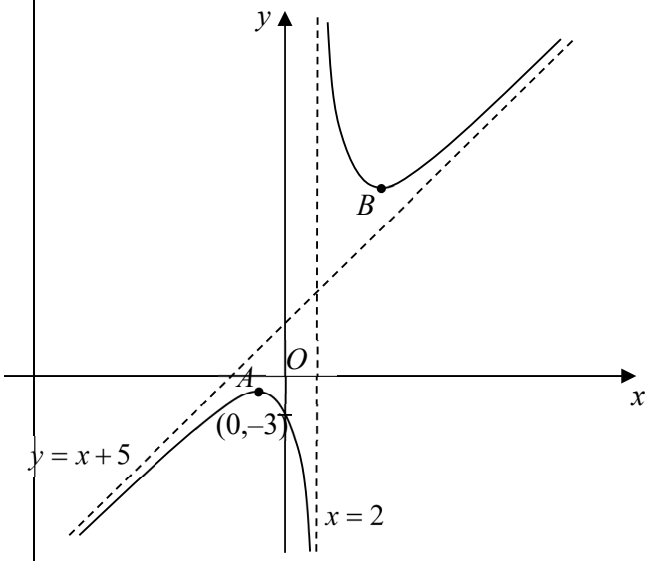
$\Rightarrow y \leq -1$ or $y \geq 15$

$\therefore y$ cannot lie between -1 and 15.

(iiii)

Graph of C:

$$y = \frac{x^2+3x+6}{x-2} = x+5+\frac{16}{x-2}$$

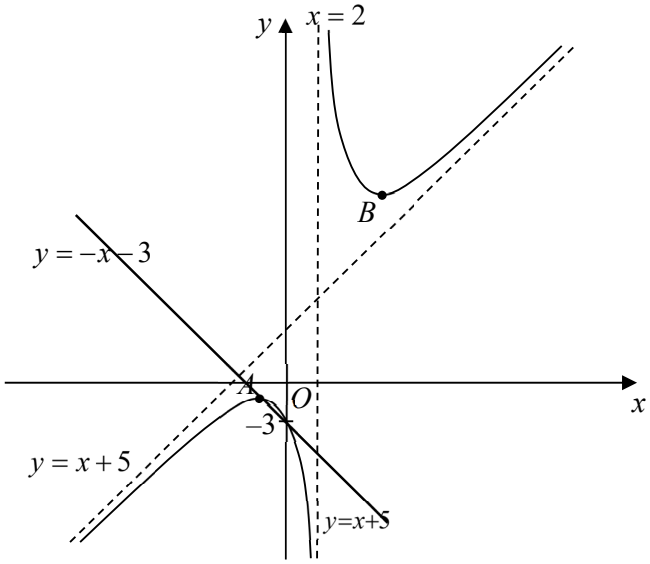


Coordinate of $A = (-2,-1)$ and $B = (6,15)$

(iv)

$$\frac{x^2+3x+6}{x-2} \geq -x-3$$

Hence, sketch additionally the graph of $y = -x-3$:



The two graphs intersect at A ($x = -2$) and at $x = 0$.

$\therefore$ Solution to $\frac{x^2+3x+6}{x-2} \geq -x-3$ is $-2 \leq x \leq 0$ or $x > 2$

7 VJC JC1 Promo 9758/2019/Q10

A curve C has equation $y = \dfrac{\alpha x^2 + x + 1}{x + 2}$, where α is a real, non-zero constant.

Show that if C has 2 stationary points, then $\alpha < 0$ or $\alpha > k$, where k is a constant to be determined. [4]

Sketch the curve C for $\alpha = -1$, giving the equations of asymptotes, the coordinates of stationary points and points of intersection with the axes. [4]

By considering C and an appropriate line, find the range of values of m such that the equation

$$-x^2 + x + 1 = mx^2 + (2m + 3)x + 6$$

has exactly 1 positive root and 1 negative root. [3]

$y = \dfrac{\alpha x^2 + x + 1}{x + 2}$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{(x+2)(2\alpha x + 1) - (\alpha x^2 + x + 1)}{(x+2)^2} = \dfrac{\alpha x^2 + 4\alpha x + 1}{(x+2)^2}$

For C to have 2 stationary points, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ has 2 real roots.

For $\alpha x^2 + 4\alpha x + 1 = 0$ to have 2 real roots,

Discrimant > 0

$(4\alpha)^2 - 4\alpha > 0$

$4\alpha(4\alpha - 1) > 0$

$\alpha < 0$ or $\alpha > \dfrac{1}{4}$

$\therefore k = \dfrac{1}{4}$

<u>Alternatively,</u>

$y = \dfrac{\alpha x^2 + x + 1}{x + 2} = \alpha x + 1 - 2\alpha + \dfrac{4\alpha - 1}{x + 2}$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \alpha - \dfrac{4\alpha - 1}{(x+2)^2}$

For C to have 2 stationary points, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ has 2 real roots.

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0 \Rightarrow \alpha - \dfrac{4\alpha - 1}{(x+2)^2} = 0$

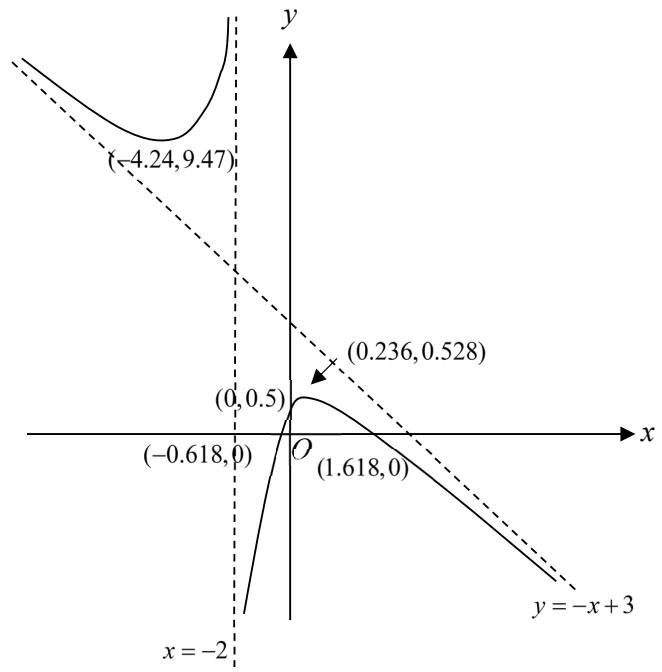
$\Rightarrow (x+2)^2 = \dfrac{4\alpha - 1}{\alpha}$

for equation to have 2 real roots,

$\frac{4\alpha - 1}{\alpha} > 0$

$\alpha < 0$ or $\alpha > \frac{1}{4}$

$\therefore k = \frac{1}{4}$



$-x^2 + x + 1 = mx^2 + (2m+3)x + 6$

$= mx(x+2) + 3(x+2)$

$\frac{-x^2 + x + 1}{x+2} = mx + 3$

Add line $y = mx + 3$ such that there are exactly 2 points of intersection with C, one with $x < 0$ and one with $x > 0$.

For the equation to have 1 positive root and 1 negative root,

$m < -1$

8 SRJC Prelims 9740/2009/01/Q7

The curve C has equation $y = \dfrac{x^2 + b}{x + a}$, where $a > 0$ and $b > 0$.

(i) State the coordinates of the intersection(s) of the curve C with the axes in terms of a and b. **[1]**

(ii) Find the equation(s) of the asymptote(s). **[2]**

(iii) Draw a sketch of the curve C, labeling the equation(s) of its asymptote(s) and coordinates of any intersection with the axes. **[2]**

(iv) Hence find the range of values of k, where k is a positive constant, for which the equation $x^2 + b = (x + a)(kx - a)$ has no real root. **[2]**

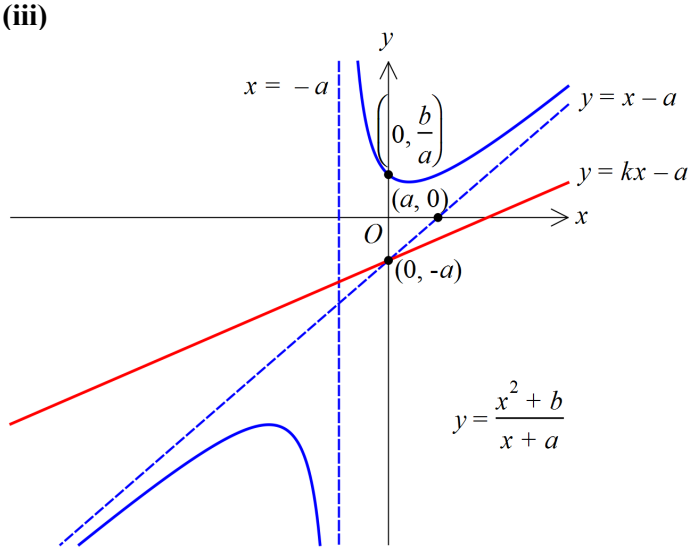
(i) $(0, \dfrac{b}{a})$

(ii) $y = \dfrac{x^2 + b}{x + a}$

$= (x - a) + \dfrac{b + a^2}{x + a}$

Equations of the asymptotes:
$x = -a$; $y = x - a$

(iii)



(iv) $x^2 + b = (x + a)(kx - a)$

$\dfrac{x^2 + b}{x + a} = kx - a$

For above equation to have no real root, the line $y = kx - a$ should not intersect C.

Since k is a positive constant, from graph, $0 < k \leq 1$

9 NYJC Promo 9740/2006/01/Q11

The curve C has equation given by $y = \dfrac{ax^2 + bx + c}{x + d}$. It has a vertical asymptote with equation $x = -2$ and its oblique asymptote cuts the x-axis at $x = -2$.

(i) State the value of d and show that $b = 4a$. **[3]**

(ii) Find the condition for a and c such that the gradient of C is always positive. **[3]**

(iii) If C passes through the origin, find the value of c. **[1]**

(iv) Given that $a = 2$ and using the values of b, c and d found in (i) and (iii), sketch the graph of C. **[3]**

(v) Find the equation of the normal line to C at the origin, and determine the coordinates of the point where this normal line cuts C again. **[4]**

Solution

(i) From the equation of C, $x = -d$ is a vertical asymptote $\Rightarrow d = 2$.

Method 1:

Since $(-2, 0)$ is on the asymptote, the asymptote has equation $y - 0 = m\left(x - \left(-2\right)\right)$ i.e.

$y = m\left(x + 2\right)$ for some m . Hence, $\dfrac{ax^2 + bx + c}{x + 2} = m\left(x + 2\right) + \dfrac{p}{x + 2}$

$$ax^2 + bx + c = m\left(x + 2\right)^2 + p$$
$$= m\left(x^2 + 4x + 4\right) + p$$

Compare coefficients of x^2 : $a = m$

$$x : \ b = 4m$$

Hence $b = 4a$ (shown).

Method 2:

$$y = \frac{ax^2 + bx + c}{x + 2}$$
$$= ax + \left(b - 2a\right) + \frac{c - 2\left(b - 2a\right)}{x + 2}$$

The equation of the oblique asymptote is $y = ax + \left(b - 2a\right)$. Since this intersects the x-axis when $x = -2$, $0 = -2a + b - 2a \ \Rightarrow b = 4a$ (shown)

(ii) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = a - \dfrac{c-4a}{(x+2)^2} = \dfrac{ax^2+4ax+8a-c}{(x+2)^2}$

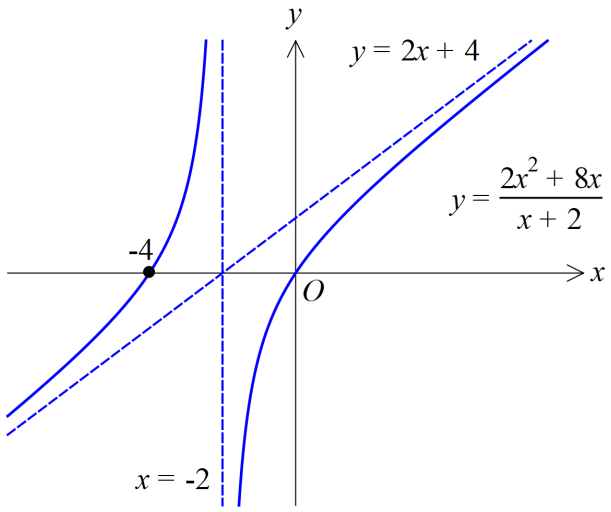
For the gradient to always be positive, since $(x+2)^2$ is always positive, the quadratic expression $ax^2+4ax+8a-c$ has to be always positive.

Hence, $a>0$ and discriminant $=(4a)^2-4(a)(8a-c)=4a(c-4a)<0$

Solving for this inequality gives us the two conditions $a>0$ and $c<4a$.

(iii) When $x=0,\ y=0.\ \therefore c=0$

(iv)



(v) $\left.\dfrac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = 4$ so the equation of the normal to the curve at the origin is $y=-\dfrac{1}{4}x$.

$$-\frac{1}{4}x=\frac{2x(x+4)}{x+2} \Leftrightarrow -x(x+2)=8x(x+4) \Leftrightarrow x(9x+34)=0$$

gives $x=0$ (original value) or $x=-\frac{34}{9}$ and $y=\frac{17}{18}$. So the coordinates of the other point of intersection is $\left(-\frac{34}{9}, \frac{17}{18}\right)$.

Conic Sections

10 VJC JC1 Promo 9758/2019/Q4

The curve C has equation $\dfrac{(y-2)^2}{9}-\dfrac{(x-3)^2}{4}=1$.

(i) Sketch C, giving the equations of its asymptotes and the coordinates of any turning points. [4]

The curve D has equation $12y^2-48y+48+ax^2-6ax-3a=0$, where a is a positive constant.

(ii) Find the set of values of a for which C and D do not intersect. [2]

(i)

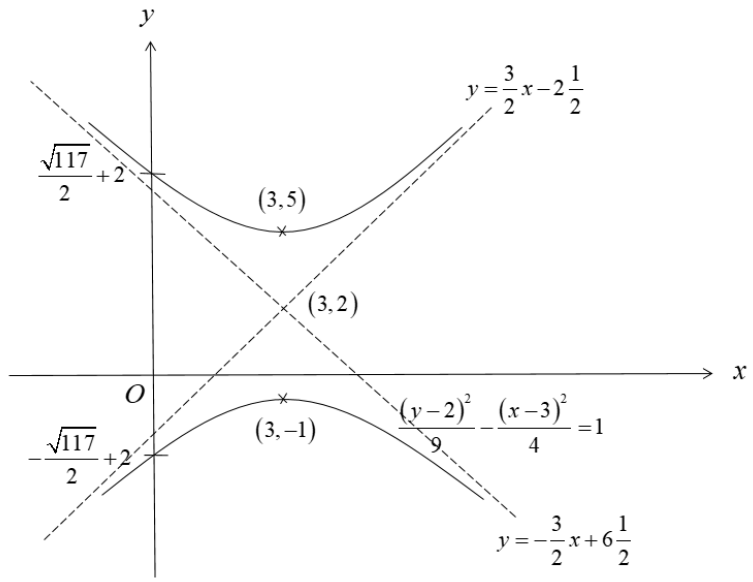
Asymptotes:

$$\frac{(y-2)^2}{9}=\frac{(x-3)^2}{4}$$

$$\frac{y-2}{3}=\pm\frac{x-3}{2}$$

$$y=\pm\frac{3(x-3)}{2}+2$$

$$y=\frac{3}{2}x-\frac{5}{2} \quad \text{or} \quad y=-\frac{3}{2}x+\frac{13}{2}$$



Note: It is optional to give the y-intercepts.

(ii)	$12y^2-48y+48+ax^2-6ax-3a=0$ $12\left(y^2-4y\right)+a\left(x^2-6x\right)-3a+48=0$ $12\left(y^2-4y+4\right)-48+a\left(x^2-6x+9\right)-9a-3a+48=0$ $12\left(y-2\right)^2+a\left(x-3\right)^2=12a$ $\dfrac{\left(y-2\right)^2}{a}+\dfrac{\left(x-3\right)^2}{12}=1$ which is an ellipse with centre $\left(3,2\right)$ and vertices at $\left(3,2+\sqrt{a}\right)$ and $\left(3,2-\sqrt{a}\right)$ For curve C and D to not intersect, $\sqrt{a}<3\Rightarrow a<9$ Since a is a positive constant, $\therefore\left\{a\in\mathbb{R}:0<a<9\right\}$

11 TMJC Promo 9758/2020/Q4

The curve C has equation $y=\dfrac{2}{x^2-k^2}$ where $k\geq 2$. It is given that C has only one stationary point.

(i) Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. Hence find the coordinates of the stationary point in terms of k.

[4]

(ii) Sketch the graph of C, stating the equations of any asymptotes, coordinates of any points of intersection with the axes and the coordinates of the stationary point. [3]

(iii) By adding a suitable graph on the same diagram in part **(ii)**, find the range of values of h, in terms of k, where h is a positive real constant and $h\neq k$ such that $\left(hx\right)^2+\left(\dfrac{2k}{x^2-k^2}\right)^2=\left(kh\right)^2$ has at least one real root.

[3]

(i)

$$y = \frac{2}{x^2 - k^2}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -2\left(x^2 - k^2\right)^{-2}(2x) = -\frac{4x}{\left(x^2 - k^2\right)^2}$$

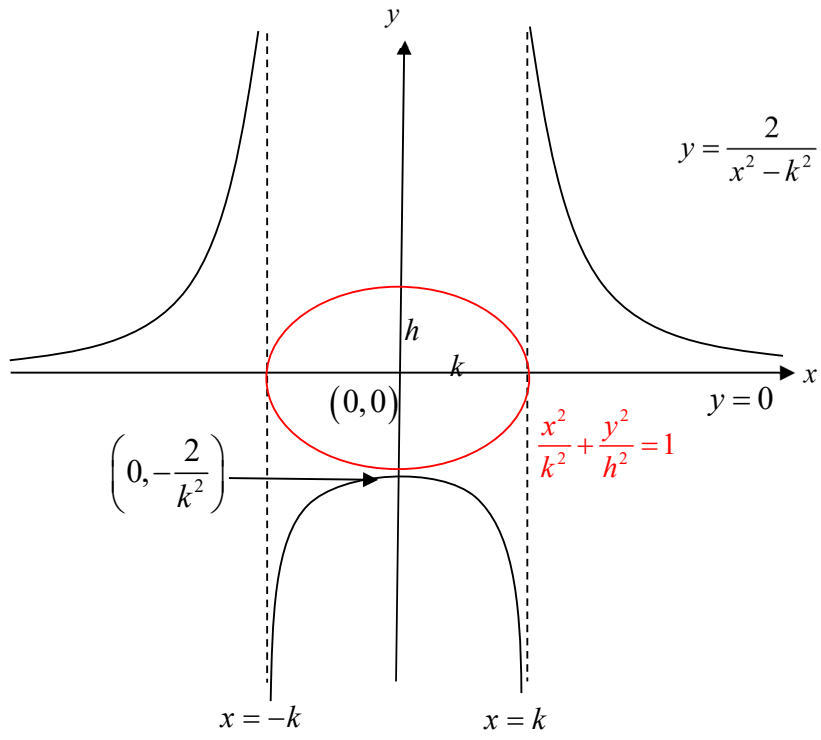
At stationary point, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

$$-\frac{4x}{\left(x^2 - k^2\right)^2} = 0 \Rightarrow x = 0$$

When $x = 0$, $y = -\frac{2}{k^2}$

$\therefore$ coordinates of stationary point is $\left(0, -\frac{2}{k^2}\right)$.

(ii)



(iii)	$(hx)^2+\left(\dfrac{2k}{x^2-k^2}\right)^2=(kh)^2$ $(hx)^2+(yk)^2=(kh)^2$ $\dfrac{(hx)^2}{(kh)^2}+\dfrac{(yk)^2}{(kh)^2}=\dfrac{(kh)^2}{(kh)^2}$ $\dfrac{x^2}{k^2}+\dfrac{y^2}{h^2}=1$ Since $h\neq k$, insert an ellipse centered at $(0,0)$ with semi-major horizontal axis k and semi-minor vertical axis h onto the graph in part **(ii)** There is at least one real solution when $h\geq\dfrac{2}{k^2}$

12 VJC Prelim 9740/2013/02/Q2(modified)

The curve C has equation $y=\dfrac{ax^2+bx+c}{x-1}$, where a, b and c are non-zero constants.

(a) It is given that C passes through the point $\left(3,\dfrac{23}{2}\right)$ and has a minimum point at $(2,10)$.

(i) Find the values of a, b and c. **[3]**

(ii) Sketch C, giving the coordinates of any turning points, points of intersection with the axes and the equations of any asymptotes. **[3]**

(iii) Find the range of values of m and k, such that the curve with equation $\dfrac{x^2}{k^2}+\dfrac{y^2}{m^2}=1$ does not intersect C. **[2]**

(b) In the case when $a=-1$ and $b=-1$, find the set of values of c, where $c\neq 2$, such that C has no stationary point. **[3]**

(ai)	$y=\dfrac{ax^2+bx+c}{x-1}$ Since C passes through the point $\left(3,\dfrac{23}{2}\right)$, $\dfrac{23}{2}=\dfrac{a(3)^2+3b+c}{3-1}\Rightarrow 9a+3b+c=23$ C passes through the point $(2,10)$ too, so $10=\dfrac{a(2)^2+2b+c}{2-1}\Rightarrow 4a+2b+c=10$ Since (2, 10) is a minimum point, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0$ when $x=2$.

$$\frac{dy}{dx}=\frac{(2ax+b)(x-1)-(ax^2+bx+c)(1)}{(x-1)^2}$$

$$=\frac{ax^2-2ax-b-c}{(x-1)^2}$$

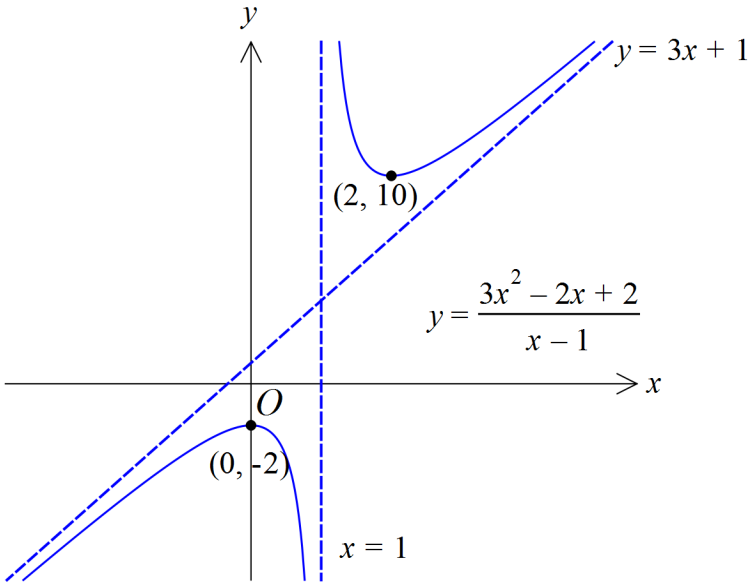
So $0=4a-4a-b-c\Rightarrow b+c=0.$

Solving the three equations using the GC, $a=3,\ b=-2,\ c=2.$

(ii)

$$y=\frac{3x^2-2x+2}{x-1}$$

$$y=3x+1+\frac{3}{x-1}$$



iii)

$\frac{x^2}{k^2}+\frac{y^2}{m^2}=1$ describes an ellipse. For the ellipse to not intersect the curve, we must have $-2<m<2,\ m\neq 0$ and $k\neq 0$.

(b)

$$a=b=-1\Rightarrow y=\frac{-x^2-x+c}{x-1}$$

(Note that when $c=2,\ y=\frac{-x^2-x+2}{x-1}=\frac{(-x-2)(x-1)}{x-1}=-x-2,\ x\neq 1$)

From (i), when $a=b=-1,\ \frac{dy}{dx}=\frac{-x^2+2x+1-c}{(x-1)^2}$

When $\frac{dy}{dx}=0,\ -x^2+2x+1-c=0$

For there to be no stationary points, the equation $-x^2+2x+1-c=0$ must have no real solutions for x. Hence, the discriminant $2^2-4(-1)(1-c)<0.$

	$4+4(1-c)<0$ $\Rightarrow c>2$ So the set of values of c is $\{c\in\mathbb{R}: c>2\}$.

Parametric Equations

13 Sketch, on separate diagrams, the curves defined by the following pair of parametric equations:

(a) $x=a\cos^3\theta,\quad y=a\sin^3\theta$ for $0\le\theta\le\pi$ where a is a positive constant.

(b) $x=t^2-4,\quad y=3t^4+8t^3$ for $t\in\mathbb{R}$;

(c) $x=1+\dfrac{1}{t},\quad y=t-\dfrac{1}{t^2}$ for $t\in\mathbb{R}, t\neq 0$

Find the equation of the curves in cartesian form. **[10]**

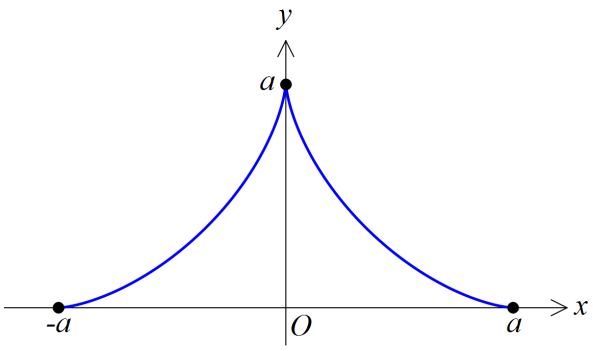
(a)

$x=a\cos^3\theta,\quad y=a\sin^3\theta$ for $0\le\theta\le\pi$,
where a is a positive constant.

$$\Rightarrow \cos\theta=\left(\frac{x}{a}\right)^{\frac{1}{3}},\quad \sin\theta=\left(\frac{y}{a}\right)^{\frac{1}{3}}$$

Since $\sin^2\theta+\cos^2\theta=1$,

$$x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}},\ y\ge 0$$

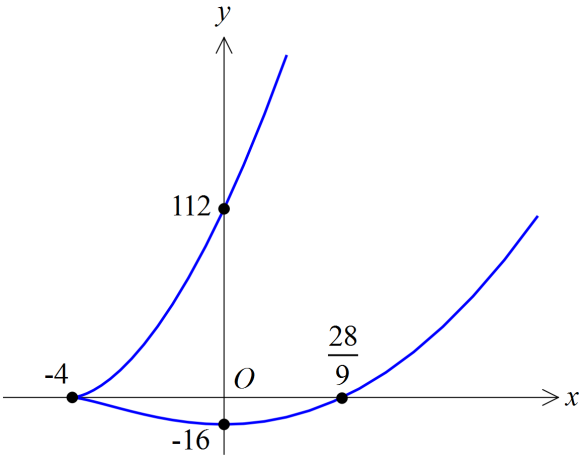


(b)

$x=t^2-4,\quad y=3t^4+8t^3$ for $t\in\mathbb{R}$

Substituting $t^2=x+4$ into $y=3t^2+8t^3$,

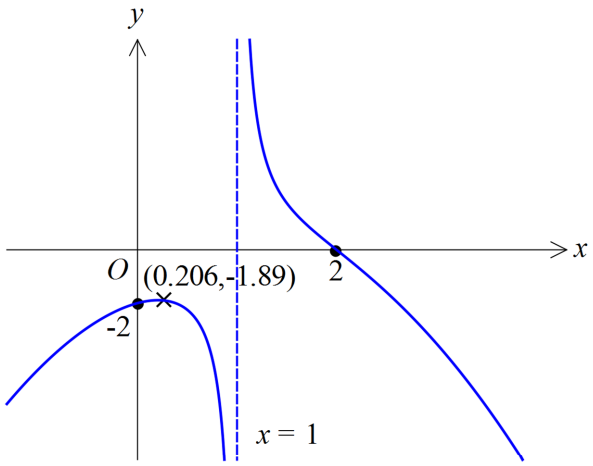
$y=3(x+4)^2\pm 8(x+4)^{\frac{3}{2}}$



(c) $x = 1 + \frac{1}{t}, \quad y = t - \frac{1}{t^2}$ for $t \in \mathbb{R}, t \neq 0$

Substituting $t = \frac{1}{x-1}$ into $y = t - \frac{1}{t^2}$,

$y = \frac{1}{x-1} - (x-1)^2$ for $t \in \mathbb{R}, t \neq 0$



14 DHS Promo 9758/2018/Q8

A curve C has parametric equations

$$x = \frac{k}{t}, \quad y = t^2 + t,$$

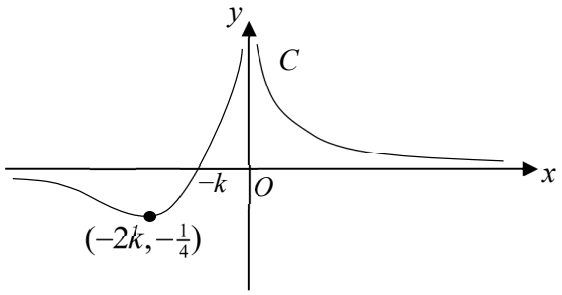
where $t \neq 0$ and k is a positive real constant.

(i) Explain why the lines $x = 0$ and $y = 0$ are asymptotes to C. [2]

(ii) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of k and t. Given that C has exactly one minimum point, show that the coordinates of the minimum point is $(-2k, -\frac{1}{4})$. [4]

(iii) Sketch C. [3]

(i)	$x = \frac{k}{t}, \quad y = t^2 + t, \quad t \in \mathbb{R}, k > 0$ As $x \to \pm\infty, t \to 0, y \to 0$. $\therefore y = 0$ is a horizontal asymptote. As $x \to 0, t \to \pm\infty, y \to \infty$. $\therefore x = 0$ is a vertical asymptote.
(ii)	$\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{k}{t^2}, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 2t + 1$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}} = \frac{2t+1}{-\frac{k}{t^2}} = -\frac{1}{k}t^2(2t+1)$ At minimum point,

	$\frac{dy}{dx} = -\frac{1}{k}t^2(2t+1) = 0$ Since $t \neq 0,\ 2t+1=0 \Rightarrow t = -\frac{1}{2}$ $x = \frac{k}{-\frac{1}{2}} = -2k,\ y = (-\frac{1}{2})^2 + (-\frac{1}{2}) = -\frac{1}{4}$ Coordinates of minimum point at $t = -\frac{1}{2}$: $(-2k,\ -\frac{1}{4})$
(iii)	When $y = 0, t(t+1) = 0 \Rightarrow t = -1\ (\because t \neq 0)$ $\therefore x = -k$ 

15 NYJC Promo/9758/2015/Q2

A curve is defined by the parametric equations

$$x = \sin t - \cos t, \quad y = \ln\left(\frac{t}{\pi}\right), \text{ where } 0 < t \leq \frac{\pi}{2}.$$

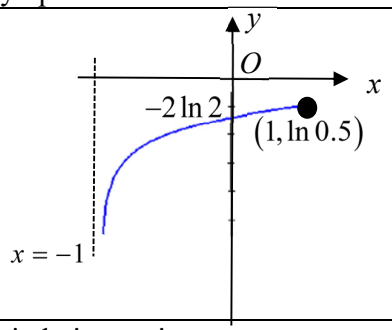
(i) Show that the gradient of the tangent to the curve at the point P, where the curve crosses the y-axis, is $\frac{2\sqrt{2}}{\pi}$. [3]

(ii) What can be said about the curve as $t \to 0$? [1]

(iii) Sketch the graph of the curve, showing clearly its endpoints if any. [2]

(iv) Hence, find the range of values of x for which the curve is strictly increasing. [1]

(i)	$x = \sin t - \cos t, \quad y = \ln\left(\frac{t}{\pi}\right), \text{ where } 0 < t \leq \frac{\pi}{2}.$ $\frac{dx}{dt} = \cos t + \sin t, \quad \frac{dy}{dt} = \frac{1}{t}$ $\frac{dy}{dx} = \frac{\frac{1}{t}}{\cos t + \sin t}$ $= \frac{1}{t(\cos t + \sin t)}$ At point P, $x = 0$

	$\sin t - \cos t = 0$ $\sin t = \cos t$ $\tan t = 1$ $t = \frac{\pi}{4}$ $\frac{dy}{dx} = \frac{1}{\frac{\pi}{4}\left(\cos\frac{\pi}{4} + \sin\frac{\pi}{4}\right)}$ $= \frac{1}{\frac{\pi}{4}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)}$ $= \frac{1}{\frac{\pi}{4}\left(\frac{2}{\sqrt{2}}\right)}$ $= \frac{1}{\frac{\pi}{2\sqrt{2}}} = \frac{2\sqrt{2}}{\pi}$ Gradient of tangent at $P = \frac{2\sqrt{2}}{\pi}$ (shown)
(ii)	As $t \to 0$, $y = \ln\left(\frac{t}{\pi}\right) \to -\infty$ $x = \sin t - \cos t \to \sin 0 - \cos 0 = -1$ $\therefore x \to -1, y \to -\infty$ $x = -1$ is a vertical asymptote.
(iii)	
(iv)	For the graph to be strictly increasing, $-1 < x < 1$

16 NJC Promo 9740/2006/01/Q3b

A curve is defined parametrically by the equations

$$x = \frac{t}{1+t}\ ,\ y = \frac{t^2}{1+t},\ t \in \mathbb{R}, t \neq -1.$$

(i) Find the Cartesian equation of the curve. **[2]**

(ii) Sketch the curve. Label your graph clearly, indicating any asymptote(s) and stationary point(s). **[3]**

(i)

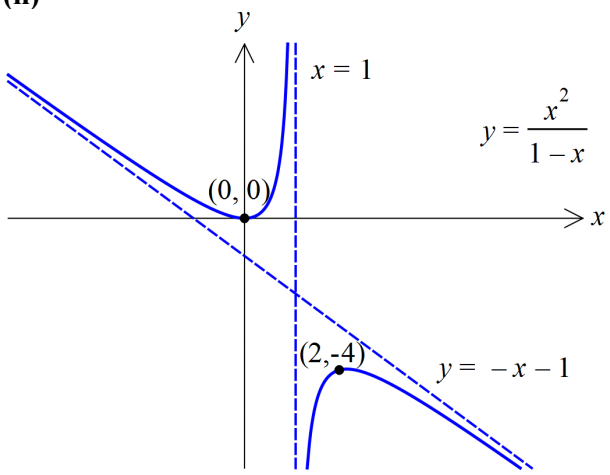
$y + x = \frac{t^2 + t}{1+t} = t \cdots (1)$

$x = 1 - \frac{1}{t+1} \Rightarrow t = -1 + \frac{1}{1-x} \cdots (2)$

Comparing (1) and (2): $y + x = -1 + \frac{1}{1-x}$

$y = -x - 1 + \frac{1}{1-x} = \frac{x^2}{1-x}$

(ii)



17 TJC Promo 9740/2006/01/Q10

A curve C is defined by the parametric equations $x = \sqrt{t},\ y = t - \frac{1}{\sqrt{t}}$, where $t > 0$.

(i) Find the coordinates of the x-intercept of C. **[2]**

(ii) By considering the behaviour of C when $t \to 0$, find the linear asymptote of C. **[1]**

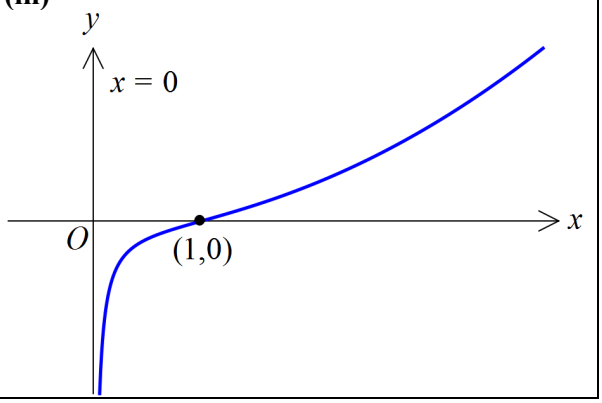
(iii) Sketch C. **[1]**

(i)
We have $y = 0$ when $t = 1$.

So x-intercept at $x = 1$.
Therefore the coordinates are (1, 0).

(ii)
As $t \to 0^+$, $x \to 0^+$ and $y \to -\infty$
The y-axis is a vertical asymptote

(iii)



18 NYJC JC1CT/9758/2020/Q1

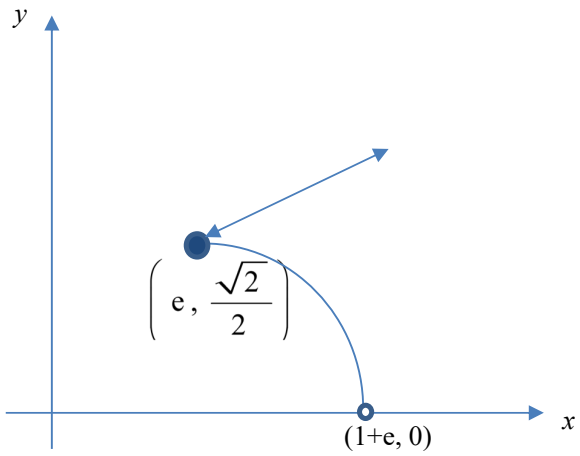
A curve C is defined by the parametric equations $x = \cos\theta + \mathrm{e},\ y = \sin\frac{\theta}{2}$ for $0 < \theta \leq \frac{\pi}{2}$.

(i) Sketch C, giving the exact coordinates of the end-points. [2]

(ii) Find the cartesian equation of C. [2]

(iii) On the same diagram as drawn in **(i)**, sketch the graph $\sqrt{2}y + 2\mathrm{e}x - y^2 - x^2 = \mathrm{e}^2 - \frac{1}{2}$. Hence find the coordinates of the point(s) where the two graphs intersect. [4]

(i)



At $\theta = 0$,

$x = \cos 0 + \mathrm{e} = 1 + \mathrm{e}$ and $y = \sin 0 = 0$

At $\theta = \frac{\pi}{2}$,

$x = \cos\frac{\pi}{2} + \mathrm{e} = \mathrm{e}$ and $y = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

(ii)

$x = \cos\theta + \mathrm{e},\ y = \sin\dfrac{\theta}{2}, 0 < \theta \le \dfrac{\pi}{2}$

$x - \mathrm{e} = \cos\theta$

$x - \mathrm{e} = 1 - 2\sin^2\dfrac{\theta}{2}$

$\sin^2\dfrac{\theta}{2} = \dfrac{1 - x + \mathrm{e}}{2}$

$y^2 = \dfrac{1 - x + \mathrm{e}}{2}$ for $0 \le y \le \dfrac{\sqrt{2}}{2}$ and $\mathrm{e} < x \le \mathrm{e} + 1$

(iii)

$\sqrt{2}y + 2ex - y^2 - x^2 = e^2 - \dfrac{1}{2}$

$y^2 - \sqrt{2}y + x^2 - 2ex = \dfrac{1}{2} - e^2$

$(y - \dfrac{\sqrt{2}}{2})^2 - \dfrac{2}{4} + (x - e)^2 - e^2 = \dfrac{1}{2} - e^2$

$(y - \dfrac{\sqrt{2}}{2})^2 + (x - e)^2 = 1$

Circle with the centre $(\mathrm{e}, \dfrac{\sqrt{2}}{2})$ and radius 1 unit.

Using GC, point of intersection at (3.60, 0.240)

